



FABRIC MANUFACTURING

Part: A

Course Code: FE 351-0723

Weft Knitting Machines

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CLASSIFICATIONS

A. According to the frame design and needle bed arrangement or construction

I. Circular knitting machine

II. Flat knitting machine

B. According to the number of needle bed or number of needle set used

I. Single Jersey knitting machine

II. Double Jersey knitting machine

C. According to the end product of the weft knitting machine

I. Fabric machine

II. Garment length machine

CLASSIFICATIONS

D. According to the basic structure of the weft knitting

I. Plain or Single jersey circular knitting machine

II. Rib Circular or Flat knitting machine

III. Interlock circular knitting machine

IV. Purl flat or circular knitting machine

E. According to the types of needle used

I. Knitting machine equipped with Latch needle

a. Circular knitting machine

One needle bed: Plain, Single-jersey jacquard, Pile and Sliver knit machine

Two needle beds (Dial-cylinder machine): Rib, Interlock, Double-jersey jacquard machine

Double cylinder: Purl knitting machine

CLASSIFICATIONS

b. Flat-bed knitting machine

One needle bed: Domestic type

Two needle beds: Vee-bed, Flat purl knitting machine

II. Knitting machine equipped with Spring Bearded needle

a. Circular knitting machine

One needle bed: Sinker wheel, Loop wheel frame

b. Straight bar frame

One needle bed and Two needle beds: Cotton's Patent or Fully-fashioned machine

TYPES OF MACHINE



S/J Circular m/c



Fleece Circular m/c



Rib Circular m/c



Interlock Circular m/c



Auto-stripe Circular m/c



S/J Jacquard Circular m/c



D/J Jacquard Circular m/c



Flat Bed Knitting m/c

FEATURES OF FLAT KNITTING MACHINE

- ❖ Flat knitting machine has *two stationary needle beds* *Latch needles are used*.
- ❖ *Angular cams of a bi-directional cam* system is used.
- ❖ The cam system is attached to the *underside of a carriage, which with its selected yarn carriers*.
- ❖ The *carriage traverses in a reciprocating manner* across the machine width.
- ❖ There is a *separate cam system for each needle bed*.
- ❖ The *two cam systems are linked together by a bridge*, which passes across from one needle bed to the other.
- ❖ Normally machine *gauge is 3 to 18 needles per inch* and machine *width up to 79 inches*.

FLAT BED KNITTING MACHINE

- ❖ The machine can be **hand-driven or power-driven**, depending on model and production requirements.
- ❖ **Fabric take-down** is achieved by rollers or weights pulling the fabric downward.
- ❖ Flat bed machines are well-suited for **fashion garments, sweaters, collars, ribs, and fully-fashioned** components.
- ❖ The structure of the machine allows **easy patterning** and **individual needle selection** by electronic or mechanical means.
- ❖ **Stitch cams** determine the depth of loop formation, and **tuck or miss cams** can be used for pattern variation.
- ❖ The machine offers **high fabric dimensional stability** and good control over fabric width and shape.

FEATURES OF FLAT KNITTING MACHINE

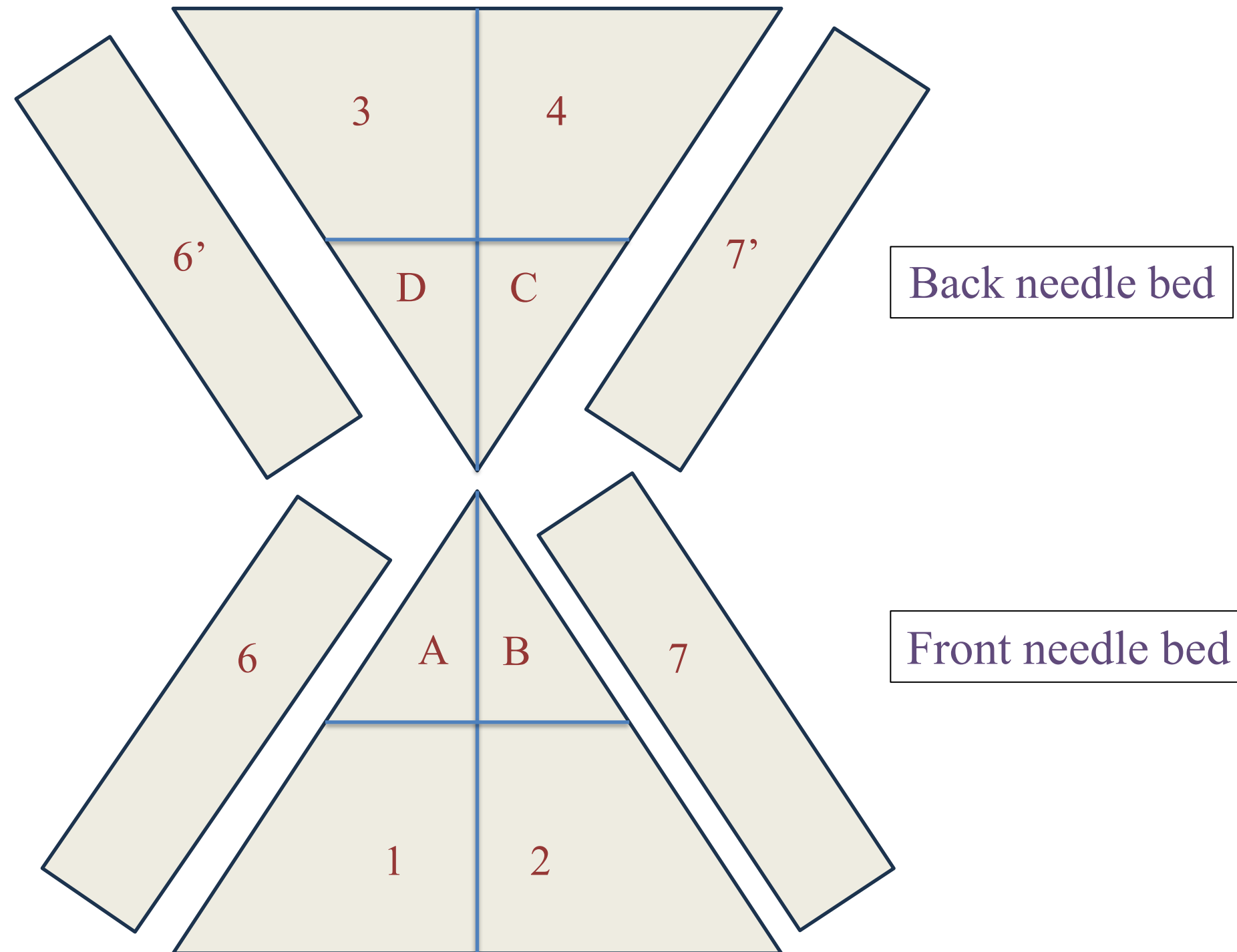


V BED KNITTING MACHINE

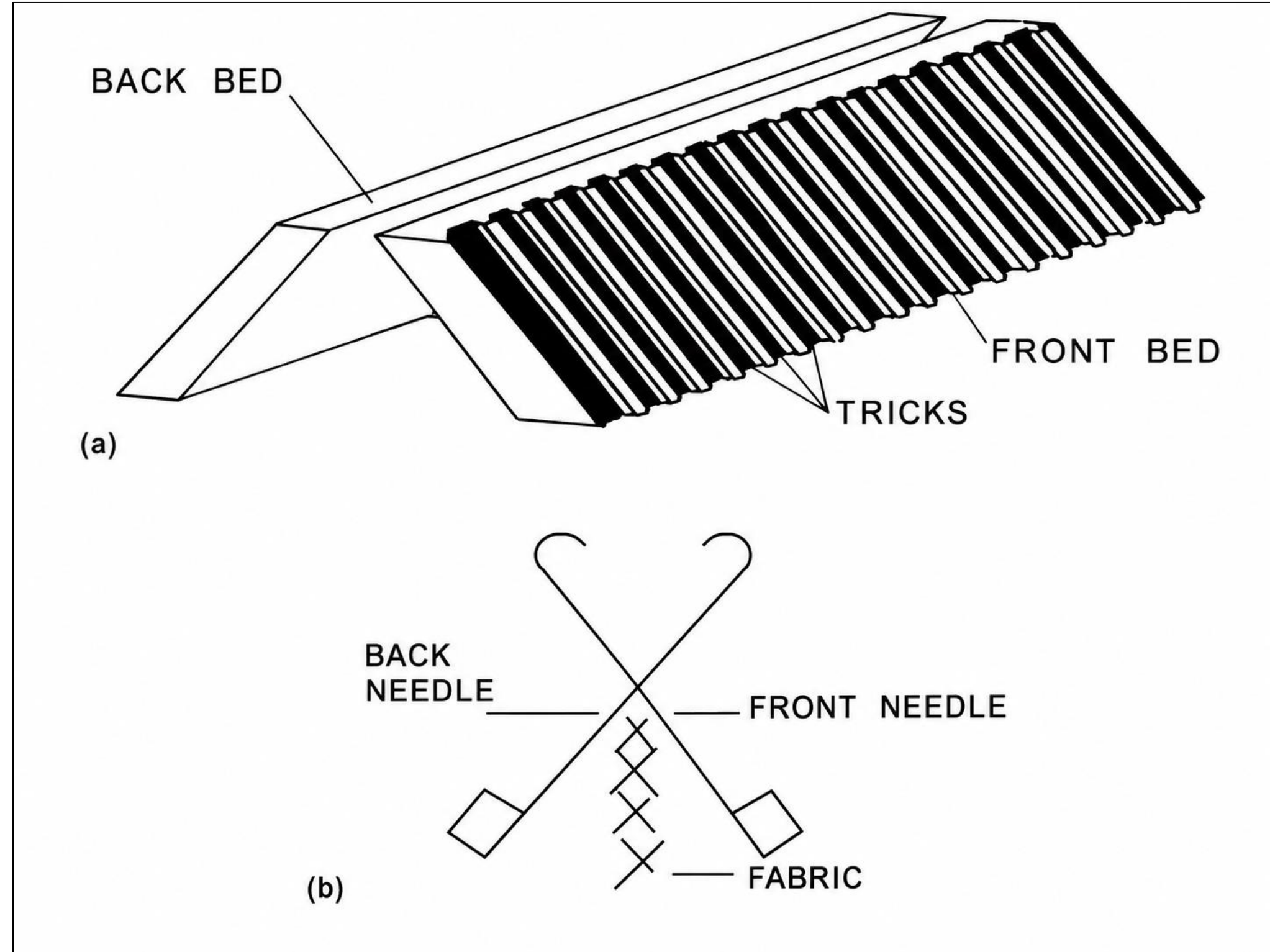
1,2,3,4 - Raising Cam

A, B, C, D - Tuck cam/Cardigan cam

6,6',7,7' - Stich cam



FEATURES OF FLAT KNITTING MACHINE



ADVANTAGES OF FLAT KNITTING MACHINE

- ❖ The flat machine is the *most versatile of weft knitting machines*, its stitch potential includes needle selection on one or both beds, racked stitches, needle-out designs, striping, tubular knitting, changes of knitting width and loop transfer.
- ❖ *A wide range of yarn counts may be knitted per machine gauge* including a number of ends of yarn in one knitting system, the stitch length range is wide and there is the possibility of changing the machine gauge.
- ❖ The operation and supervision of the machines of the simpler type is relatively *less arduous than for other weft knitting machines*.
- ❖ The number of garments or panels simultaneously knitted across the machine is dependent upon its knitting width, yarn carrier arrangement, yarn path and package accommodation.

CLASSIFICATIONS OF FLAT KNITTING MACHINE

There are four main types of flat knitting machines:

1. Vee-bed Flat Knitting Machine

It is the most common type of flat knitting machine. It has two needle beds arranged in a V-shape. It is widely used for making: Sweaters, Collars, Cuffs, Other knitted garments. It has a wide range of patterns and shaping possibilities.

2. Flat Purl (Links-Links) Knitting Machine

This machine uses double-headed latch needles. It works in a similar way to the Vee-bed machine. It has: Good pattern-making ability. Better shaping options. Lower production speed compared to some other machines. It is used for different types of knitted fabrics. Both hand-operated and automatic versions are available.

CLASSIFICATIONS OF FLAT KNITTING MACHINE

3. Domestic Single-Bed Flat Knitting Machine

These machines are mainly used for home knitting, not industrial production. They have only one needle bed. Needles are moved by cams attached to a carriage that is moved by hand. Pattern making and shaping are controlled manually. A typical machine has: About 36-inch working width. 5 - 6 needles per inch.

4. Uni-directional Multi-carriage Flat Knitting Machine

This type is made by only one manufacturer. It uses multiple carriages moving in one direction. It is designed for special knitting operations.

FEATURES OF CIRCULAR KNITTING MACHINE

1. Circular knitting machine has normally rotating (clockwise) cylindrical needle bed.
2. On circular knitting machines mainly latch needles are used.
3. For single jersey machine holding down sinkers are used, one between every needle space. For double jersey machine, dial needle bed replaces the sinker.
4. Normally stationary angular cam systems are used for needle and sinker.
5. Latch needle cylinder and sinker ring or dial revolve through the stationary knitting cam systems.
6. For single jersey machine, sinker ring which is simply and directly attached to the outside top of the needle cylinder.
7. Stationary yarn feeders are situated at regular intervals around the circumference of the rotating cylinder.
8. Yarn is supplied from cones placed either on an integral over head bobbin stand or on a free-standing creel through tensioners, stop motions and yarn guide eyes down to the yarn feeder guides.

CLASSIFICATION OF CIRCULAR KNITTING MACHINE

There are *three main types* of circular knitting machines:

A) Revolving Cylinder Latch Needle Machines

These machines produce most weft knitted fabrics.

They have two main types:

1. Open Top / Sinker Top / Single Jersey Machine

- ☐ It has one set of needles arranged in the cylinder.
- ☐ It is mainly used to produce single jersey fabrics.
- ☐ It may have pattern-making ability or may not.

2. Dial and Cylinder Machine

- ☐ It has two sets of needles: Cylinder needles (vertical) & Dial needles (horizontal)
- ☐ These machines produce: Rib fabrics, Interlock fabrics.
- ☐ They may have patterning systems.

CLASSIFICATION OF CIRCULAR KNITTING MACHINE

B) Revolving Cylinder Bearded Needle Single Jersey Machine

There are two types:

1. Sinker Wheel Machine
2. Loop Wheel Frame

Both types have similar features:

- Needles are fixed in the needle bed.
- The needle bed rotates during knitting.
- Special parts move the yarn and loops along the needles.
- The fabric is produced in a tube shape.
- The technical back side of the fabric faces outside.
- They use fewer feeders.
- Their production speed is lower, but they can make: Special knitted structures. High-quality fabrics.

CLASSIFICATION OF CIRCULAR KNITTING MACHINE

C) Circular Garment Length Machines

- ☐ These machines are used to produce complete garment lengths (large pieces of knitted fabric).
- ☐ They are usually body-width size or larger.
- ☐ They may have: Cylinder and dial arrangement, or Double cylinder arrangement.
- ☐ Small diameter versions are used for hosiery products (such as socks).
- ☐ Hosiery machines may have:
- ☐ Single cylinder, Cylinder and dial, or Double cylinder.

FEATURES OF FABRIC LENGTH MACHINE

- Fabric machines are circular knitting machines that produce *continuous tubular fabric of fixed width*.
- They are *large diameter latch needle machines* that work at *high speed*.
- After the required length is knitted, the fabric is removed manually, usually in roll form.
- Most fabrics are produced on single cylinder (single jersey) and cylinder-dial (double jersey) machines because they have high production speed and efficiency.
- *Tubular fabric may need to be opened into flat width fabric by splitting, unless a tubular shape is required.*
- After knitting, the fabric is finished and cut into garments or used for household and technical purposes.
- Fabric machines differ in productivity, flexibility, and pattern-making ability.
- Usually, cam settings and needle arrangements are not changed during fabric production.

FEATURES OF GARMENTS LENGTH MACHINE

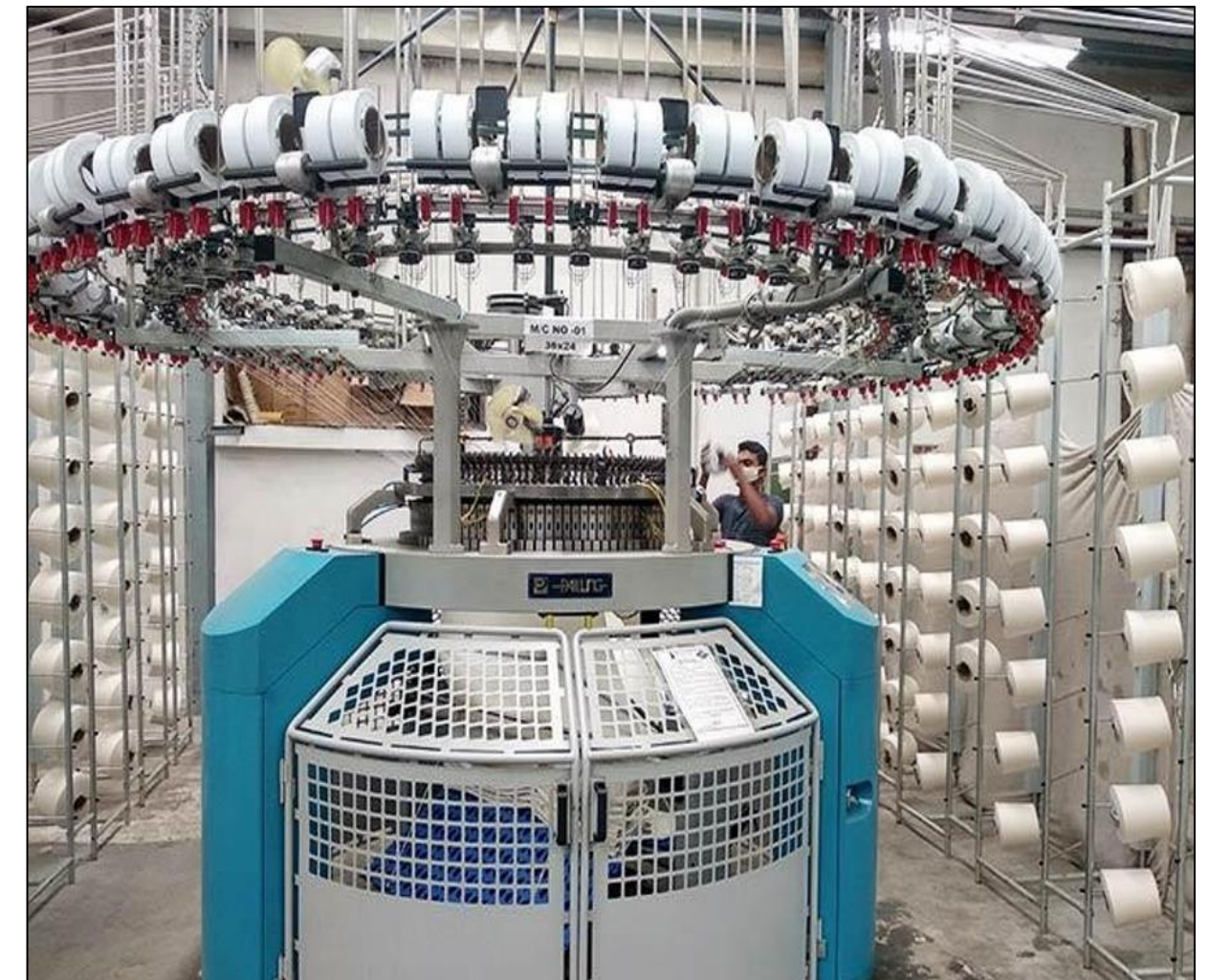
- Garment-length machines include *flat machines, hosiery machines, glove machines, straight bar frames, and circular garment machines.*
- They *produce complete garment-length pieces instead of continuous fabric.*
- These machines use a control system with timing and counting devices to manage the garment knitting sequence.
- The garment width may change during knitting according to the design.
- They *generally have a coarser gauge than fabric machines.*
- The take-down system is more advanced because it must handle different production speeds and finished garment removal.
- Garments can be knitted: In tubular form, or In open width form.
- They are used to produce: Knitwear (sweaters, pullovers, cardigans), Outerwear, Underwear.

FEATURES OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

1. It is a single-bed circular knitting machine having only one set of needles.
2. The needles are latch-type and mounted vertically in the slots of the needle cylinder.
3. The sinkers are fitted between every two needles and move radially.
4. The machine produces plain fabric (single jersey) which has technical face and back sides.
5. Usually, the diameter of the machine ranges from 12 to 60 inches.
6. The gauge may vary from 14 to 36 needles per inch, depending on yarn count and end use.
7. Cam systems are placed around the cylinder to control needle movement for knit, tuck, and miss stitches.
8. The yarn feeders (generally 90-120 per machine) feed yarns continuously to each knitting head.
9. Sinker ring which is simply and directly attached to the outside top of the needle cylinder.
10. The fabric is drawn down through the center of the cylinder by take-down rollers.
11. The production rate depends on machine diameter, speed (revolutions per minute - RPM), and number of feeders.
12. The machine frame holds the main drive, take-down mechanism, fabric collection, and lubrication systems.
13. Fabrics produced are used for T-shirts, underwear, sportswear, and other lightweight garments.

SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

- Creel
- Cone package holder
- Yarn passing tube
- Tensioner
- Positive feeder
- VDQ pully
- Stop motion device
- Feeder
- Needle
- Cam
- Cam box
- Cylinder
- Sinker
- Fabric spreader
- Take down roller
- Cloth roller
- Lubrication system
- Dust removing fan

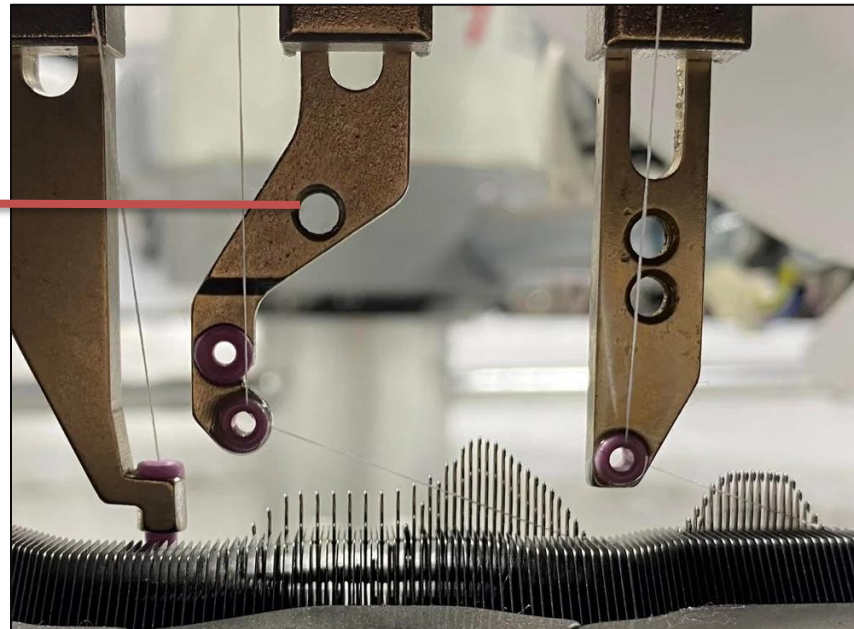


SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

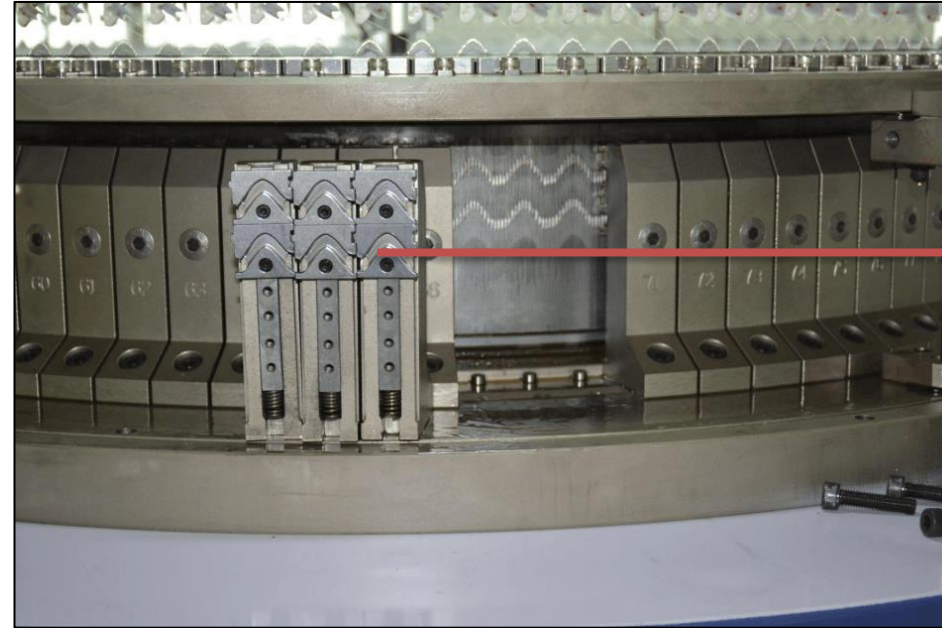


SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

YARN FEEDER



CAM



CYLINDER



SPREADER



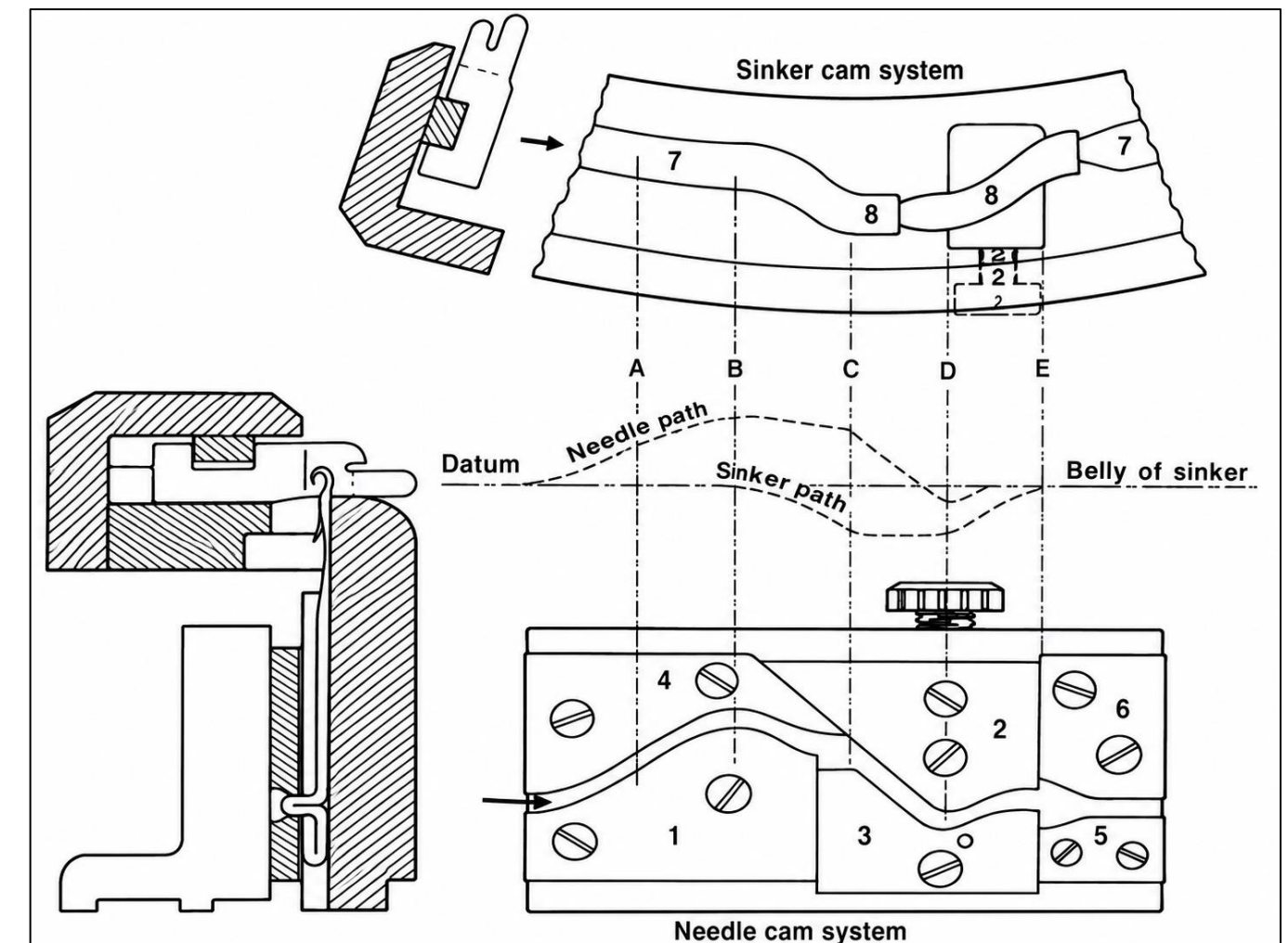
CAM SYSTEM OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

The cam system has two main parts: Needle cam system & Sinker cam system.

The needle cam race consists of six main cams:

1. Clearing cam (Raising cam)
2. Stitch cam (Lowering cam)
3. Upthrow cam.
4. Guard cam of clearing cam.
5. Return cam brings the needle back to its position.
6. Guard cam of return cam.

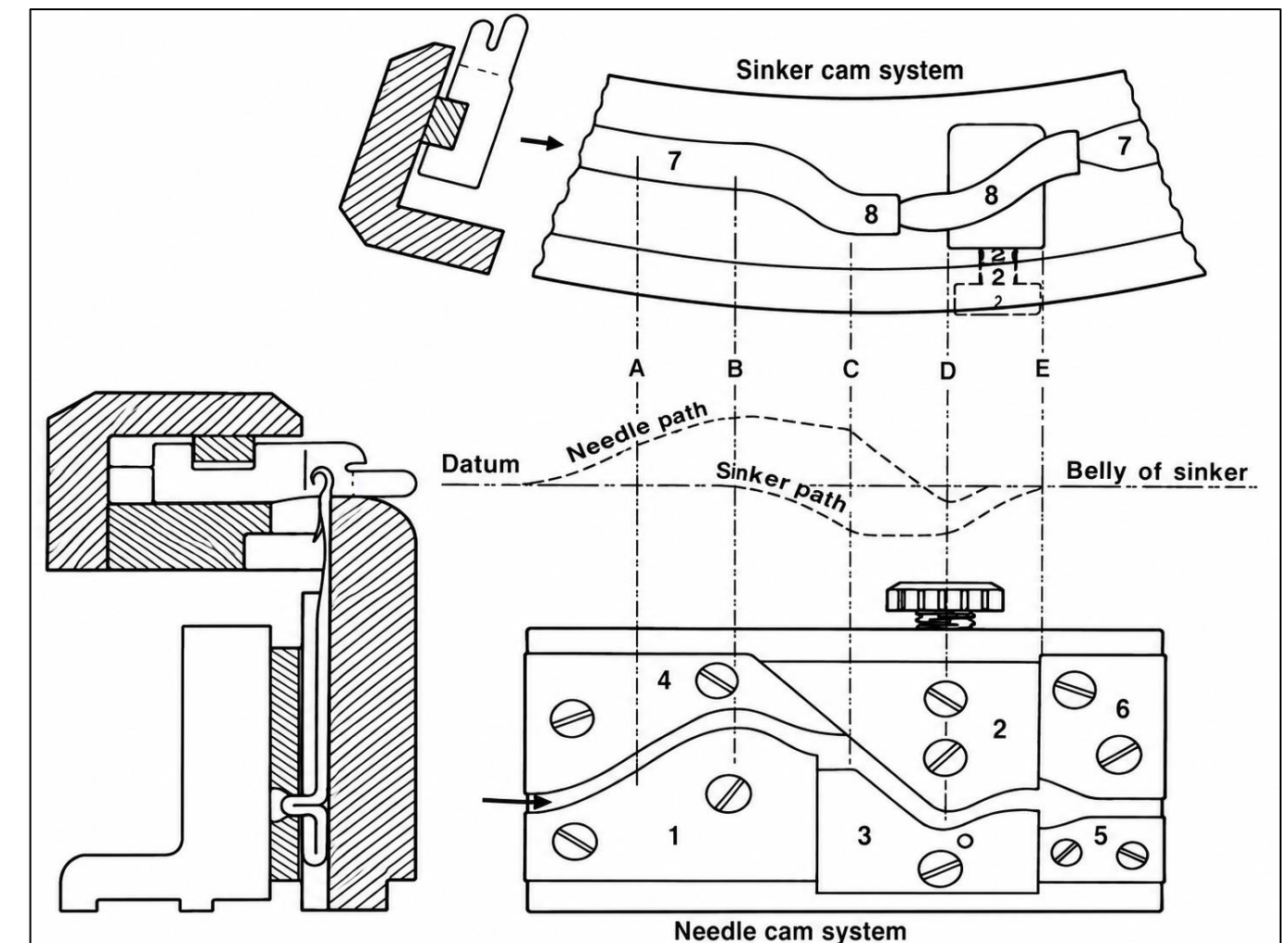
The stitch cam (2) and upthrow cam (3) can be adjusted vertically to change the stitch length.



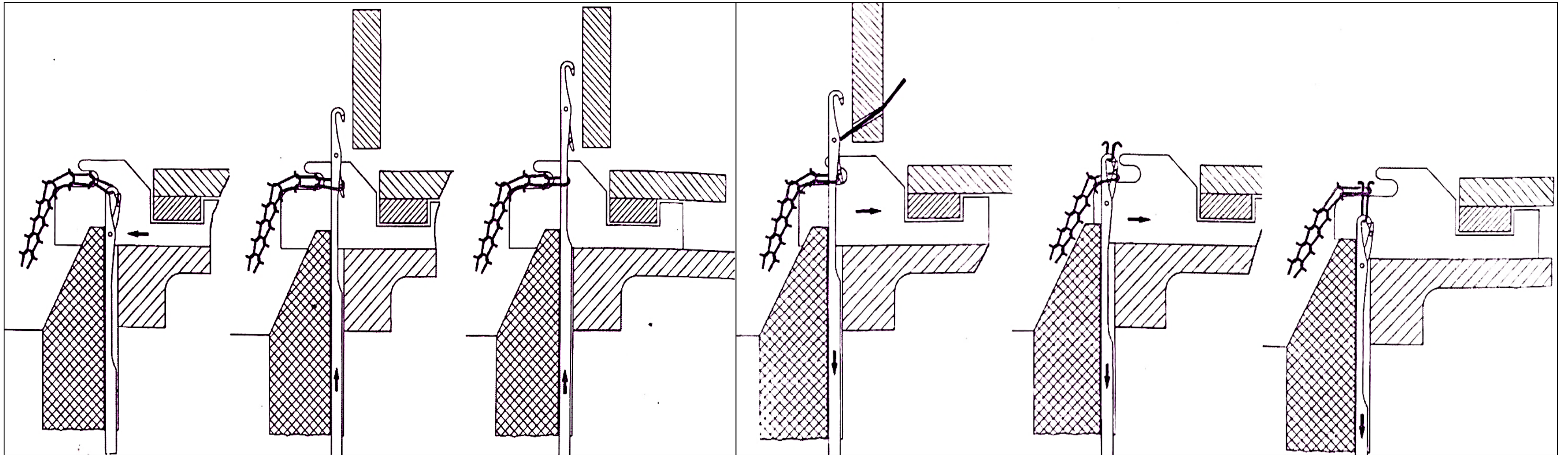
CAM SYSTEM OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

The sinker cam race consists of the followings:

- 7. The race cam
- 8. The sinker- withdrawing cam and
- 9. The sinker- return cam.



KNITTING ACTION OF SJCKM



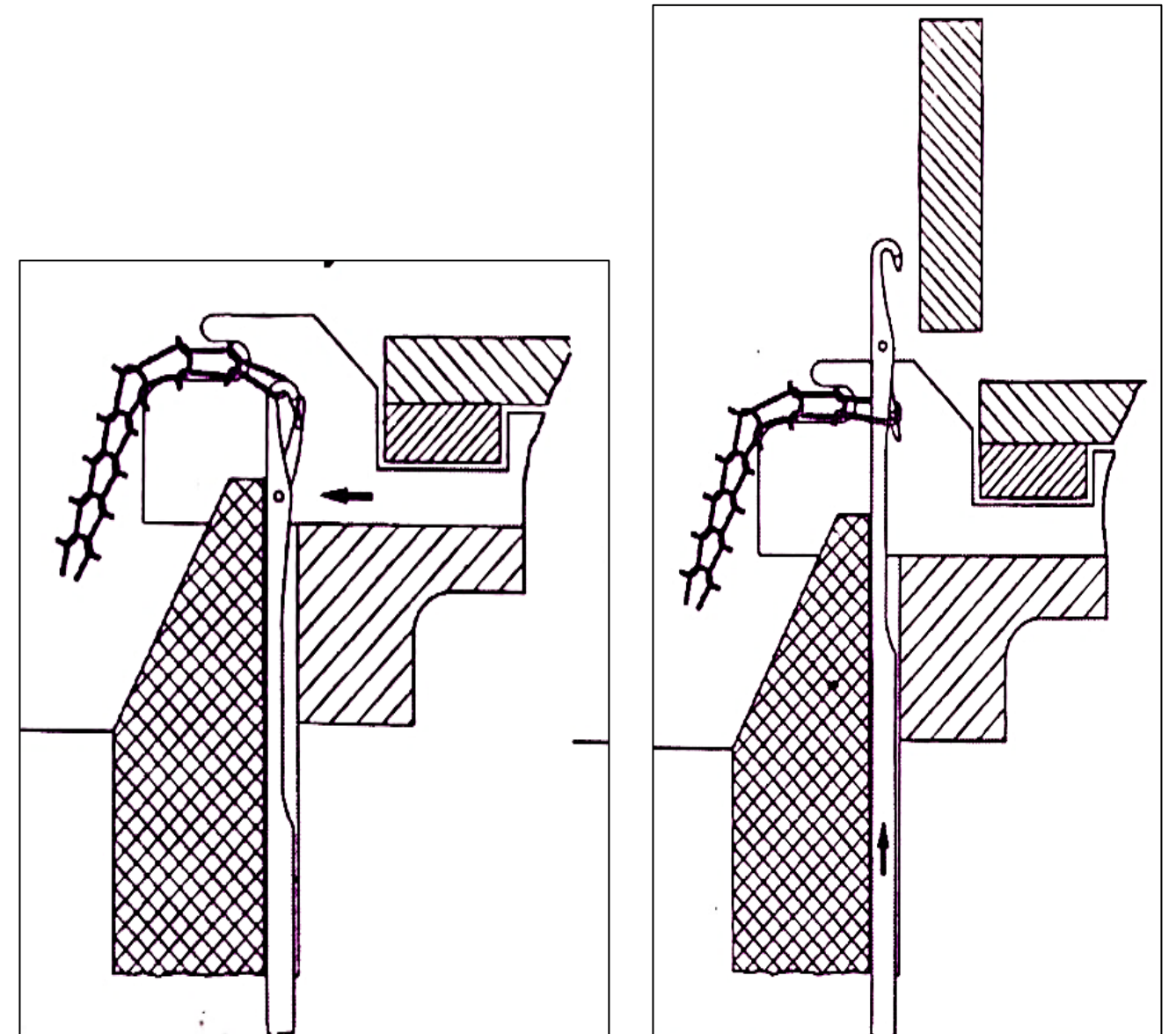
KNITTING ACTION OF SJCKM

Position 1: Rest Position

In the rest position, the *needle is at its starting point*. The top edge of the needle head is *level with the knock-over edge of the sinker*. The *sinker moves towards the center of the cylinder* and its *throat holds the old loop* and keeps the fabric in position.

Position 2: Tucking-in Position

In this position, the *needle moves upward from the rest position*. During this movement, the *old loop opens the needle latch and rests on the latch*. The yarn feeder keeps the latch open and prevents it from closing. *The sinker does not move during this stage*.



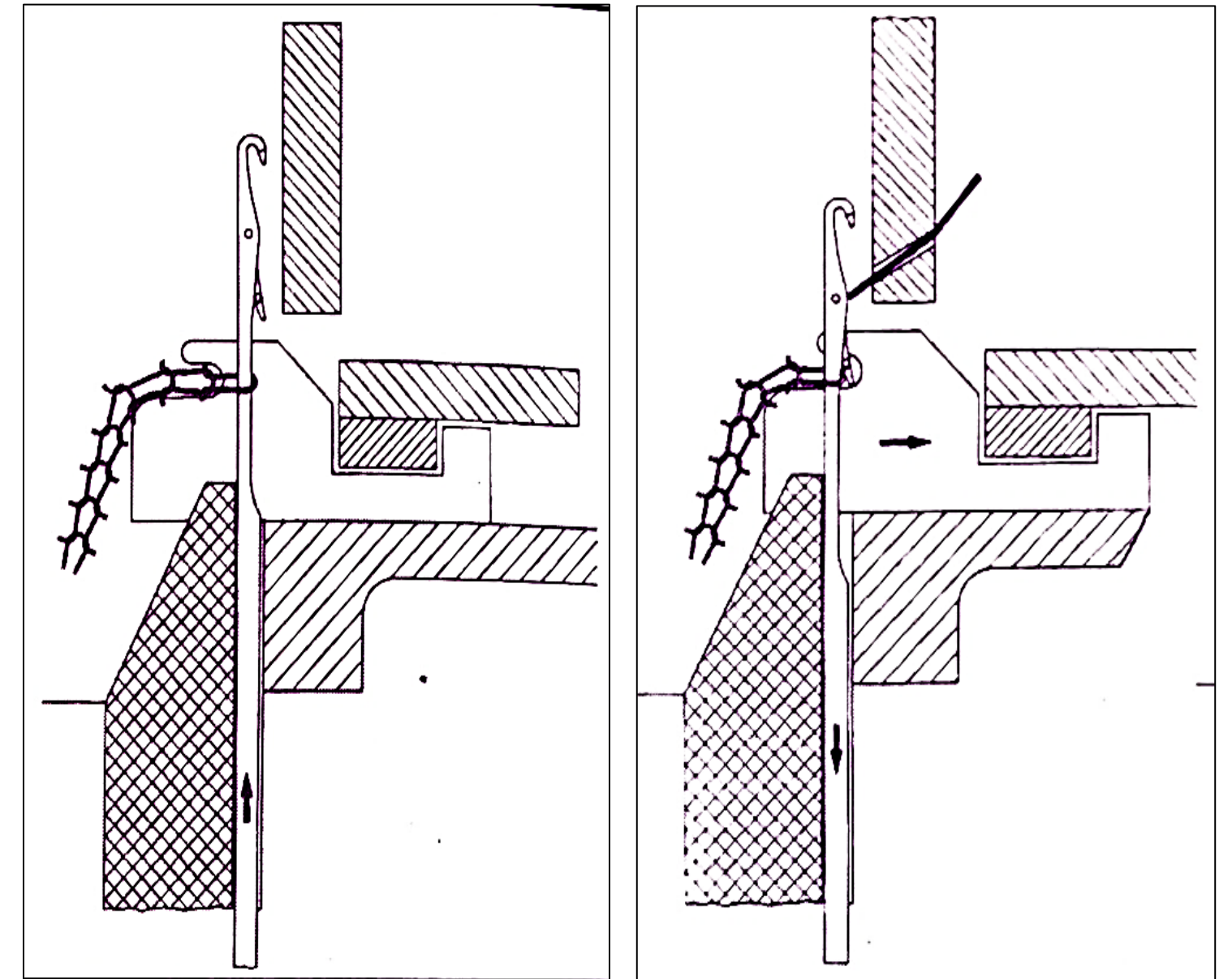
KNITTING ACTION OF SJCKM

Position 3: Clearing Position

The needle moves further upward and reaches its highest position. *Since the sinker holds the fabric down, the old loop slides along the needle stem and moves below the latch.* The yarn feeder prevents the latch from closing, and the sinker remains still.

Position 4: Yarn Presenting Position

The *needle starts moving downward from the clearing position.* Before the old loop closes the latch, *the yarn feeder places the new yarn inside the needle hook.* After the yarn is fed, *the latch can close during further downward movement.* At this stage, *the sinker moves away from the cylinder center because the fabric no longer needs to be held by the sinker throat.*



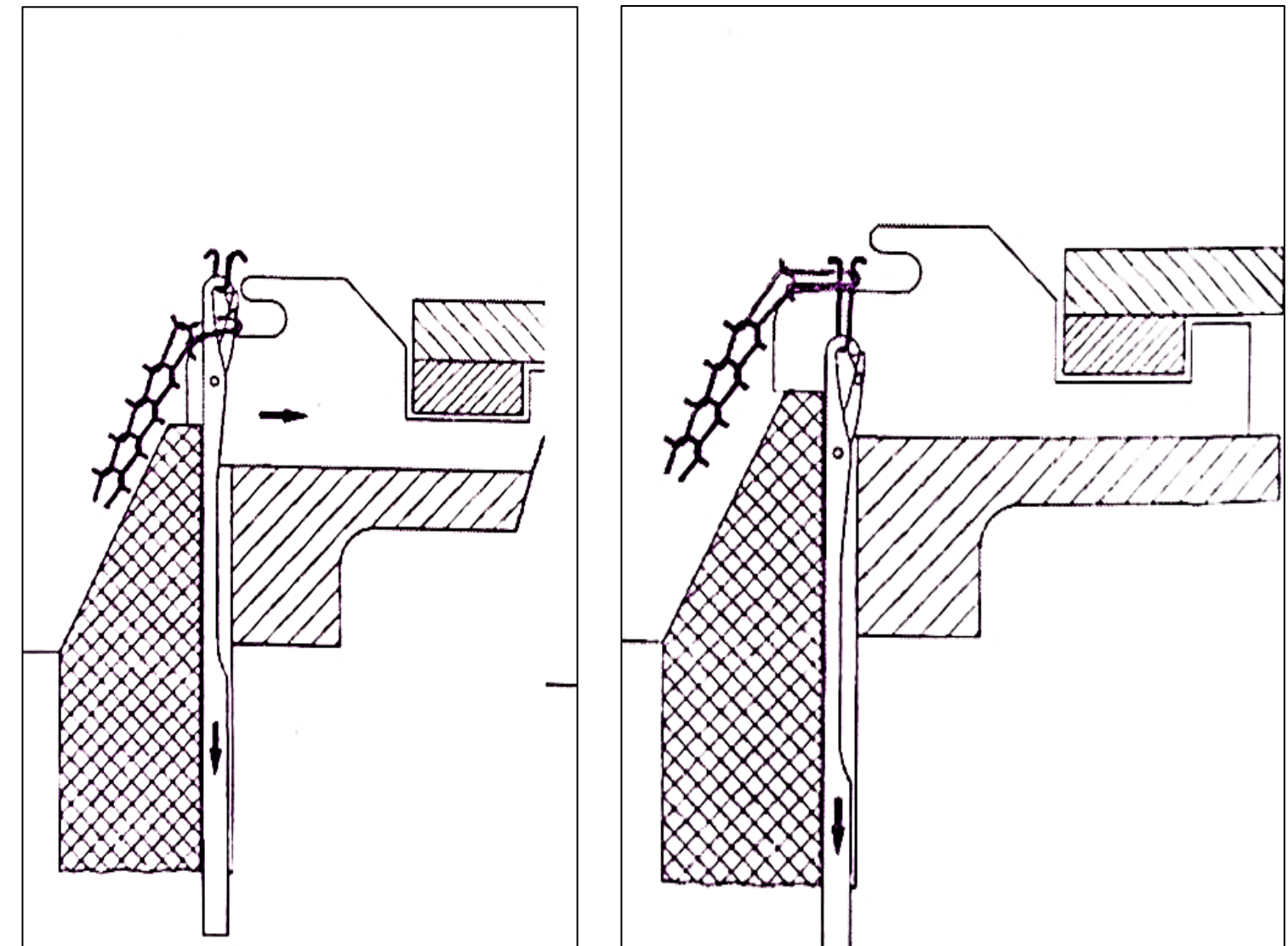
KNITTING ACTION OF SJCKM

Position 5: Cast-in Position

The needle continues moving downward from the yarn presenting position. The old loop closes the needle latch, and *the new yarn is trapped inside the closed needle hook*. The sinker reaches its extreme position and *holds the old loop at the knock-over edge so that the new yarn can be pulled through it*.

Position 6: Knocking over Position

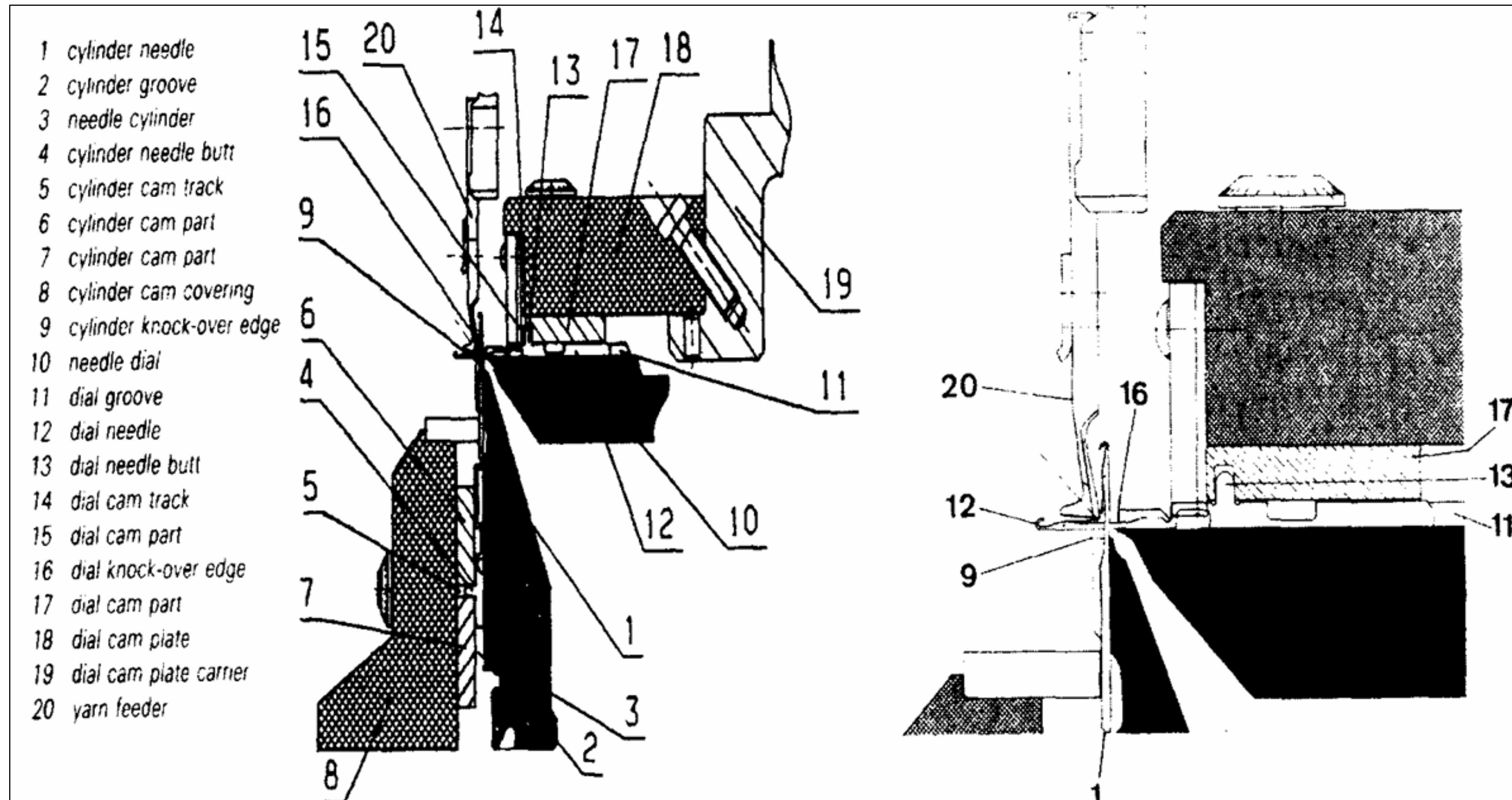
The needle moves further downward and pulls the new yarn through the old loop. *The old loop slips off the needle and a new stitch is formed*. The movement of the take-down and guide segments controls the length of the newly formed loop, allowing accurate adjustment of stitch size.



FEATURES OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE

- In this machine group there is one set of needles on the circumference of a vertical cylinder and a second set of needles, arranged perpendicular to the first set and mounted on a horizontal dial.
- On most of the circular knitting machines the cylinder and dial rotate, whereas the cams with yarn feeder guides are stationary.
- The following figure shows a cross-sectional view of the region containing the knitting elements of a rib circular knitting machine. The set-up of the cylinder (3) with its knitting elements (1 to 9) is the same as with plain circular knitting machines. In a horizontal dial (10) grooves (11) are milled in. The latch needles (12) are housed and guided in these grooves. The dial needle (12) obtains its motion for stitch formation through its butt 13, which extends into a cam track (14). This cam track (14) is formed by the cam parts (15) and (17), which in turn are fixed to a dial cam plate (18). During the rotation of the cylinder and the dial the cylinder needle (1) is moved vertically and the dial needle 12 is moved horizontally, corresponding to the shape of the cam track in the cylinder and dial cams.
- In a gauge range from 5 to 20 needle per inch.

FEATURES OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE



NEEDLE GAITING

The way of two beds of knitting needles and their housings are *aligned with each other in a dial and cylinder* of circular knitting machine or flat bed knitting machine.

Generally, two types of needle gaiting are used. These are:-

- Rib gaiting
- Interlock gaiting

NEEDLE GAITING

1. Rib Gaiting: When the needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed **at alternate position** with each other is known as Rib gaiting.



Fig: Rib Gaiting

NEEDLE GAITING

2. Interlock Gaiting: When needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed at **face-to-face position** with each other is known as Interlock gaiting.

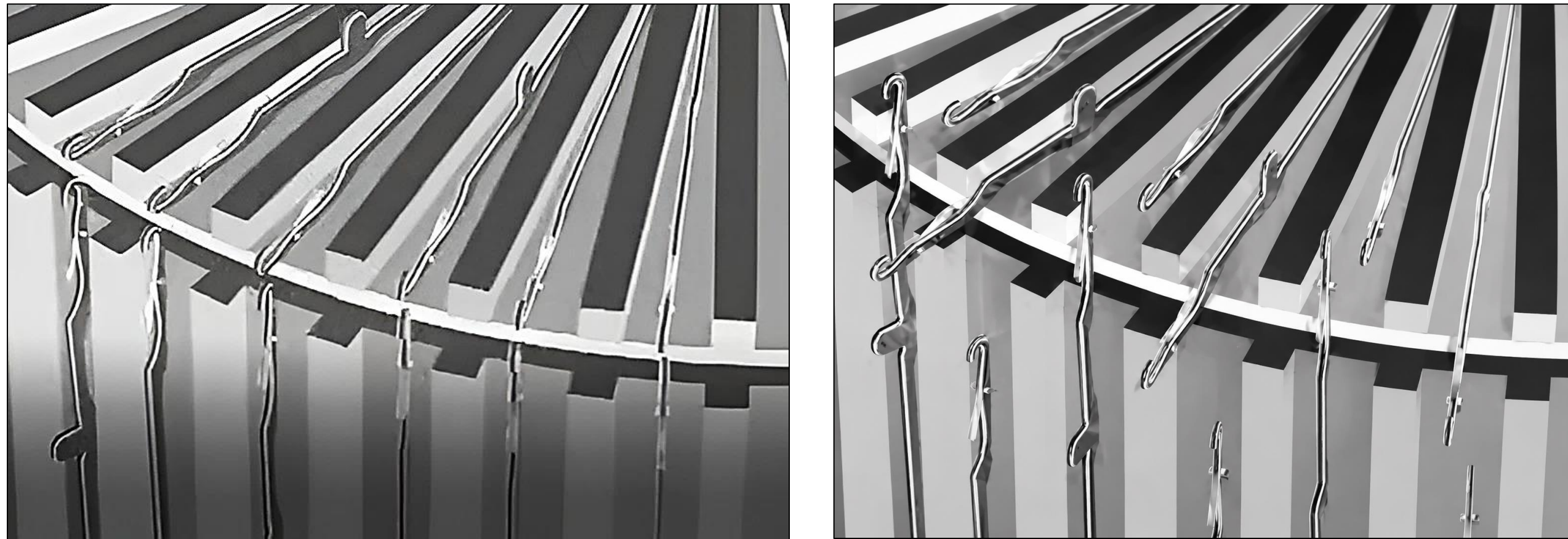
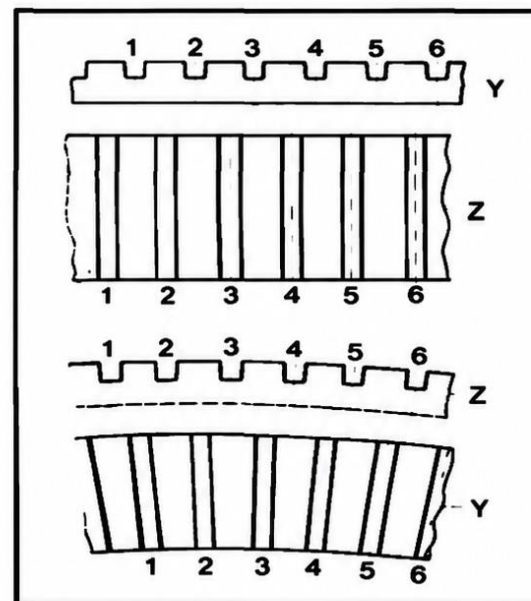
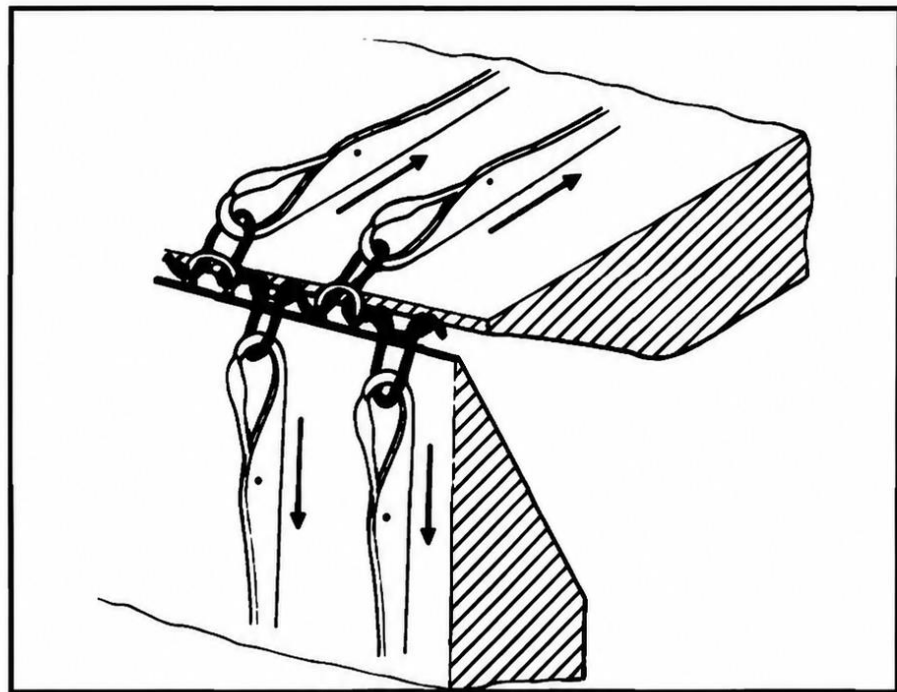


Fig: Interlock Gaiting

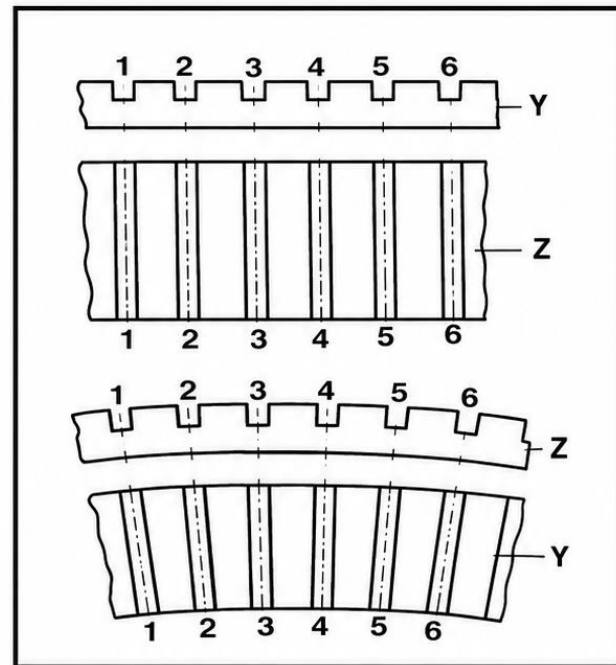
NEEDLE GAITING



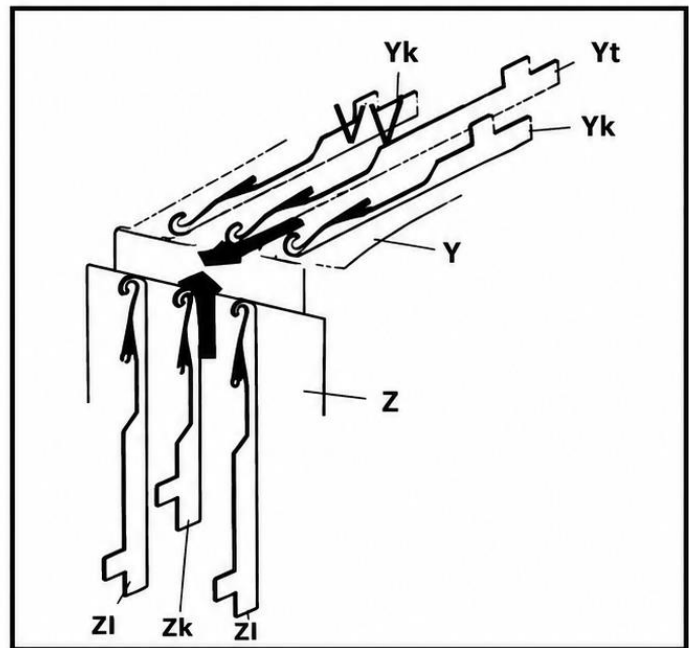
Rib gaiting for cylinder and dial on a rib circular knitting machine in front view (above) and top view (below)



Rib position of dial and cylinder needles



Interlock gaiting for cylinder and dial on a rib circular knitting machine in front view (above) and top view (below)



Interlock position of dial and cylinder needles

CONVERSION FROM RIB TO INTERLOCK GATING

- ❑ A rib knitting machine can be changed into an interlock machine if it has different types of needles in both the cylinder and dial, and the needles can be controlled separately.
- ❑ To convert it, the cylinder or dial is rotated by half a needle pitch so that the cylinder and dial needles come directly opposite each other.
- ❑ After this change, the machine works like a normal interlock knitting machine.
- ❑ In theory, an interlock machine can also be changed into a rib machine, but this is not common because interlock machines usually have finer gauges.
- ❑ If cylinder and dial needles cross too closely in rib settings, there may not be enough space, which can cause yarn damage or machine problems.
- ❑ Generally, interlock circular knitting machines use fine gauges (usually finer than E24).

NEEDLE TIMING

Needle timing is the *position difference between the dial needle knock-over point and the cylinder needle knock-over point* in a circular knitting machine.

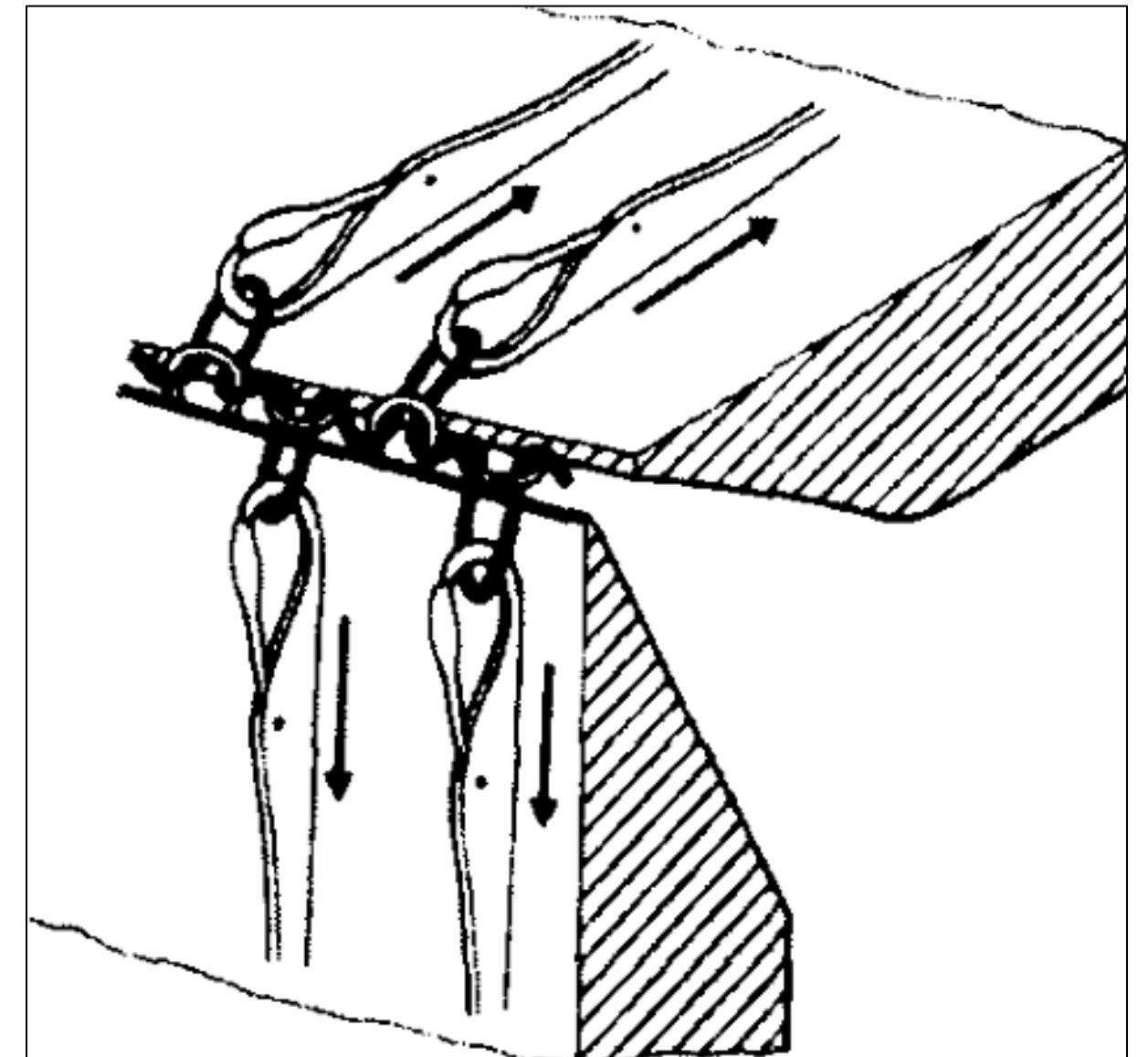
It is adjusted by changing the position of the dial cam plate compared with the cylinder cam. Individual adjustment can be made by changing or replacing the stitch cam profile.

There are two types of needle timing:

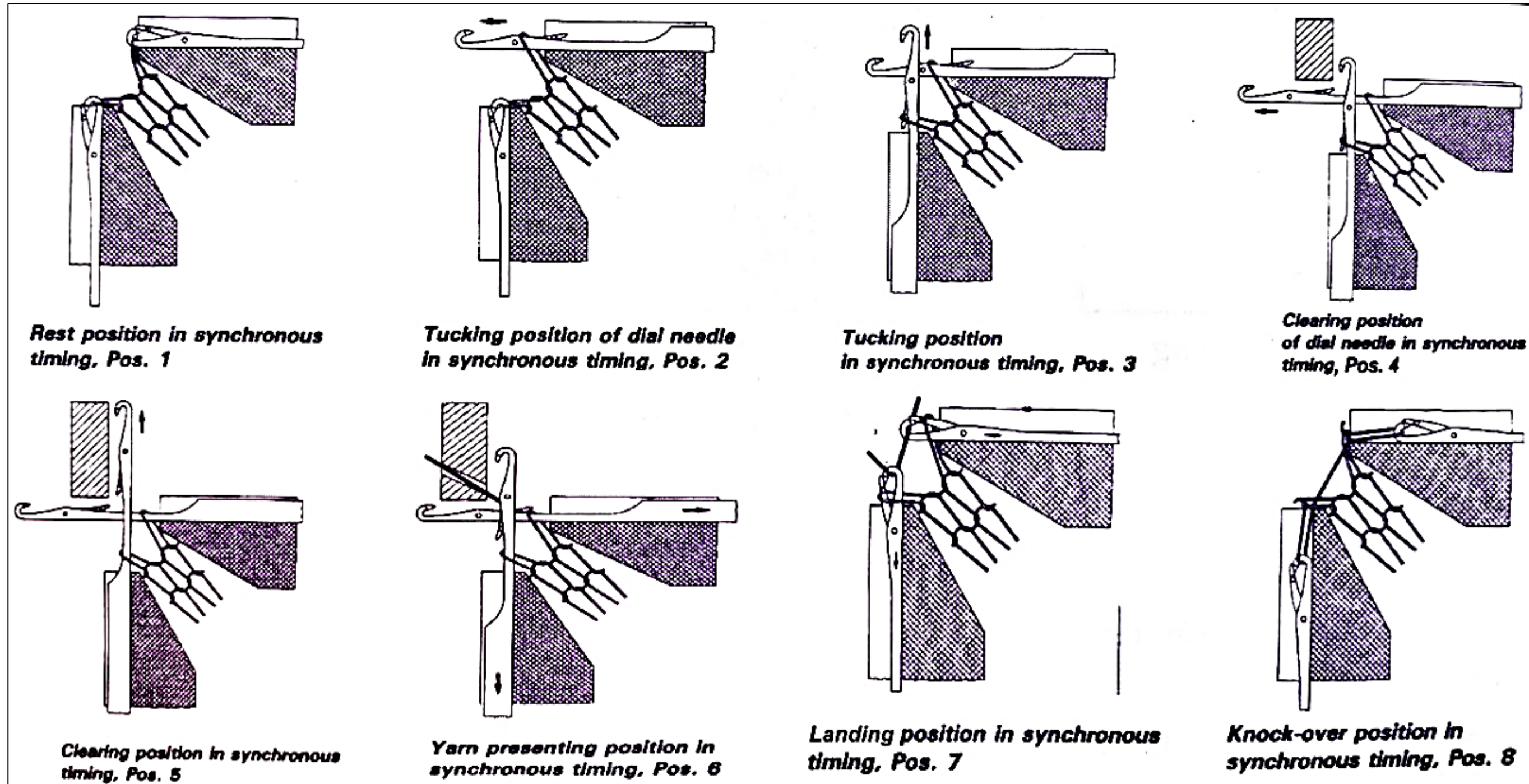
- ❑ Synchronized timing: Dial and cylinder needles knock over at the same time (also called point or 2×2 timing).
- ❑ Delayed timing: Dial needle knock-over occurs later than the cylinder needle (used for rib or interlock fabrics).

SYNCHRONIZED TIMING

- ❖ In synchronized timing, the cylinder needles and dial needles knock-over their loops at the same time.
- ❖ Both needles reach the knock-over point together, and their knock-over depths are equal.
- ❖ During loop formation, the yarn is pulled in two opposite directions by the cylinder and dial needles, which creates high tension in the yarn.
- ❖ This timing can be used for most rib and interlock structures.
- ❖ Fabrics produced with synchronized timing are usually looser and have uneven stitches because of the high yarn tension.

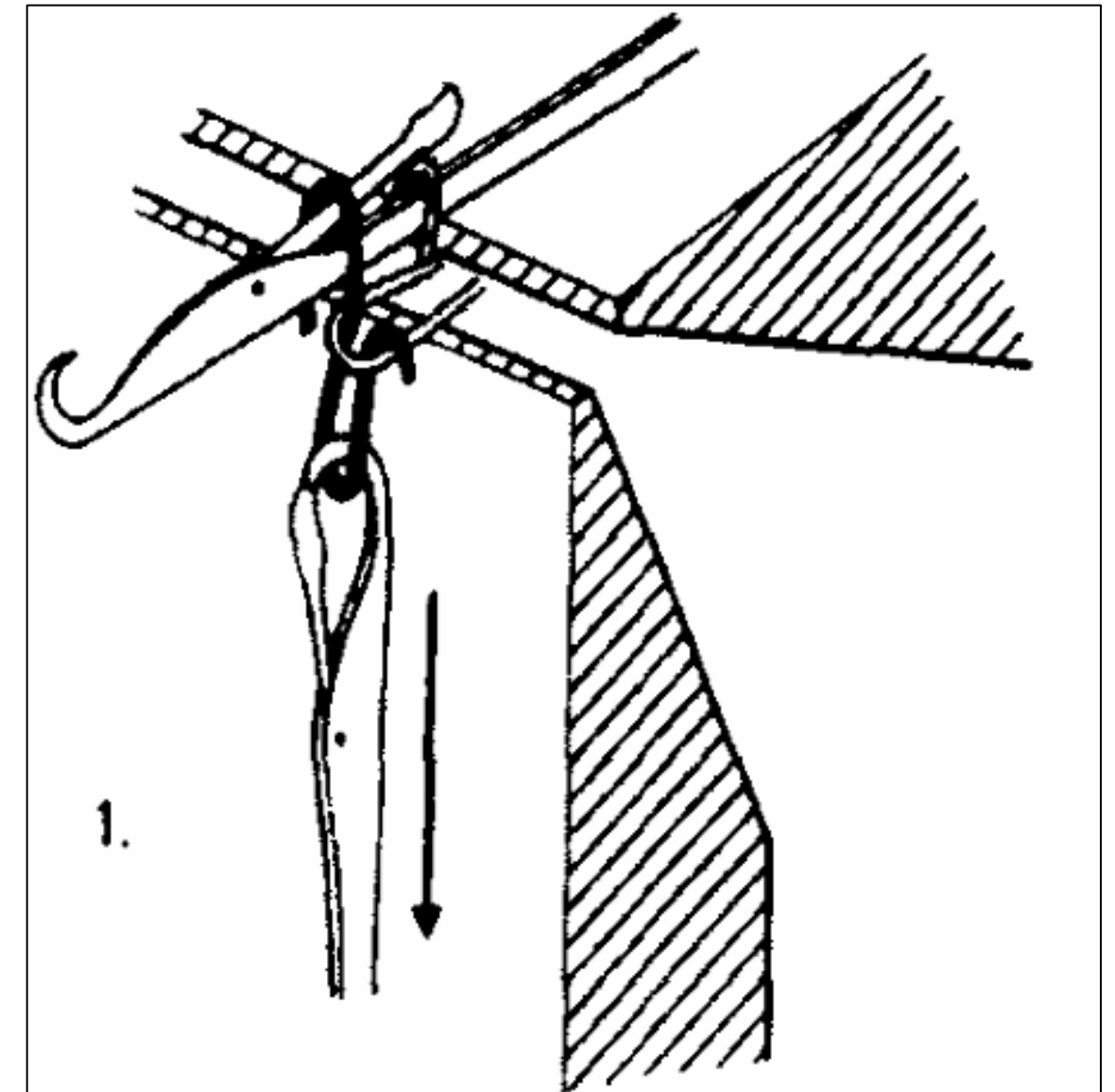


KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN SYNCHRONOUS TIMING

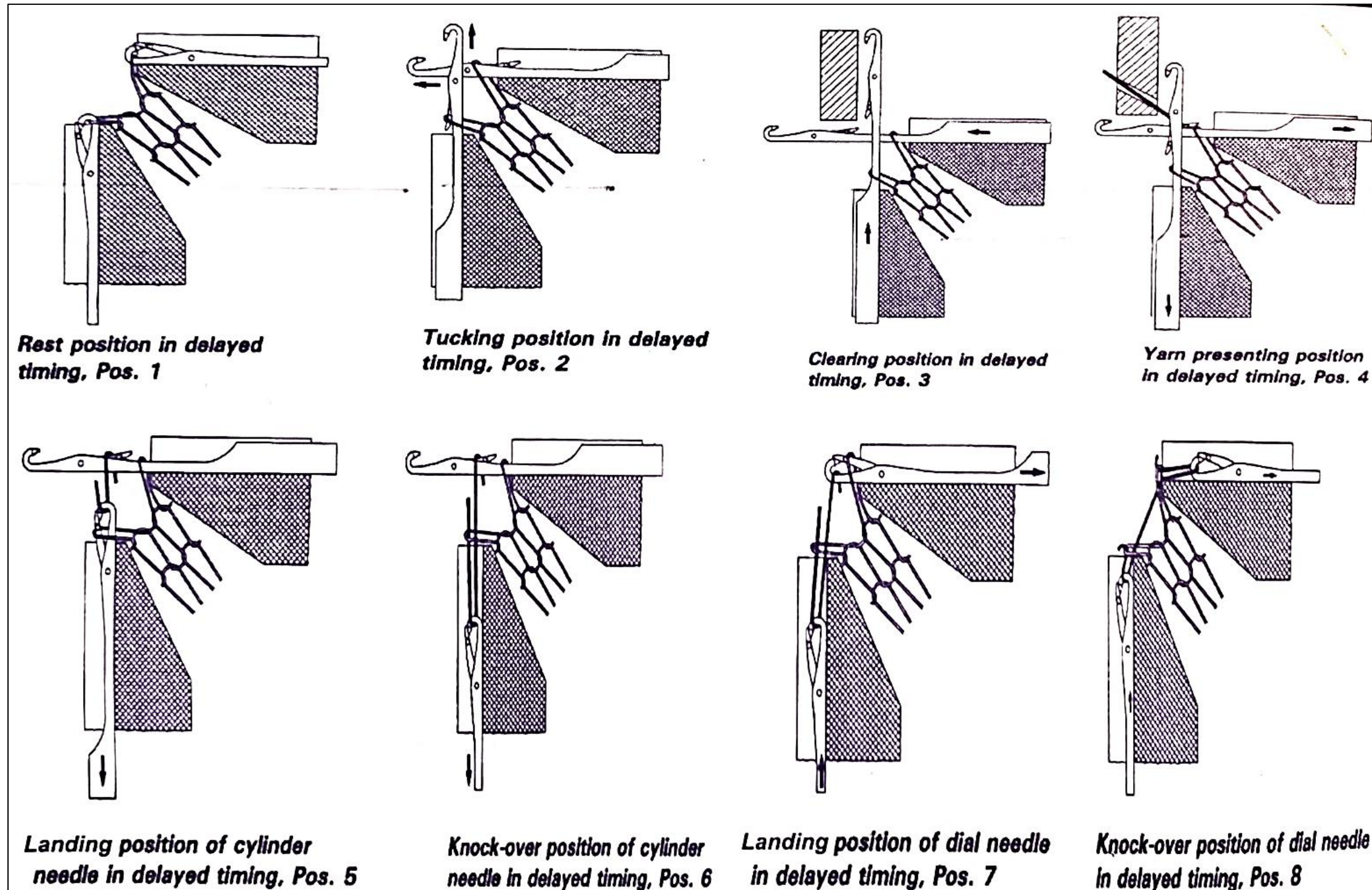


DELAYED TIMING

- ❖ In delayed timing, the dial needles knock-over their loops after the cylinder needles.
- ❖ The cylinder needles first pull the loops and then rise slightly to reduce yarn tension. After this, the dial needles complete their knock-over process.
- ❖ Because of this delay, the dial loops become longer than the cylinder loops.
- ❖ The fabric produced with delayed timing is:
 - ✓ Tighter in structure, even in stitch formation
 - ✓ More rigid and stable, heavier and wider
 - ✓ Produced with less strain on the yarn



KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN DELAYED TIMING



DELAYED TIMING

Why are dial loops longer?

Because the dial needles have extra yarn available. During delayed timing: The cylinder needles have already drawn some yarn. The dial needles then pull their loops and receive a small amount of extra yarn ("robbed" from cylinder loops). Therefore, the dial loop length becomes larger.

Then how are stitches more even?

Normally, in synchronized timing: Both needle sets pull the yarn at the same time. **The yarn tension is very high** and unequal forces can create irregular loops.

In delayed timing: Cylinder and dial needles work separately instead of pulling against each other at the same moment. **The yarn tension is reduced.** The loops form more smoothly and consistently. So, even stitches do not mean cylinder and dial loops are the same size. It means the stitches are more regular and stable.

INTERLOCK CIRCULAR KNITTING MACHINE

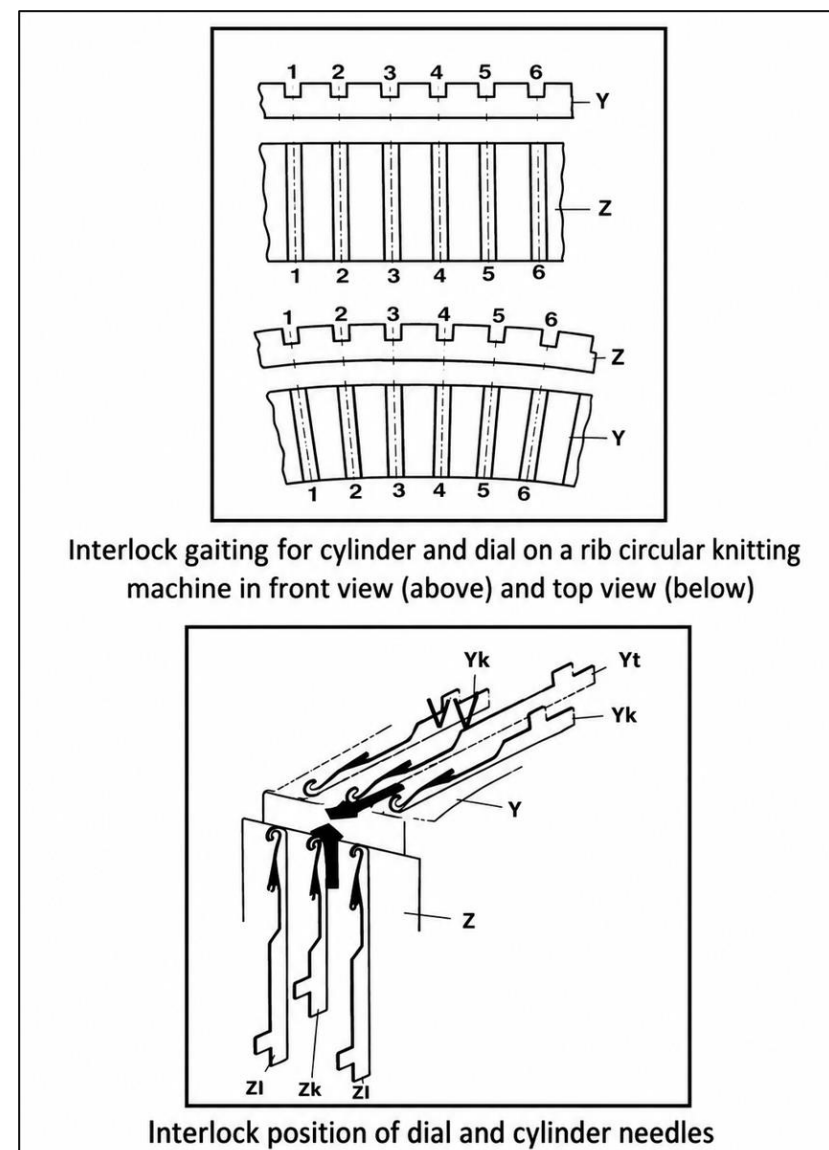
Interlock fabric is mainly produced on special cylinder and dial circular machines and some double-system Vee-bed flat machines.

Requirements of an Interlock Machine:

The *needles in the cylinder and dial must be exactly opposite each other*. This is called interlock gating. At each feeder, only one set of needles works, while the other set follows a non-knitting path. Each needle bed has two separate cam systems: One controls half of the needles at one feeder. The other controls the remaining needles at the next feeder. Needles are arranged in an alternate pattern, so opposite needles do not knit together at the same feeder. Conventional Interlock Machine uses two types of needles: Long needles & Short needles. Long needles work at the first feeder. Short needles work at the second feeder. Needles are arranged alternately, with a long needle opposite a short needle.

INTERLOCK CIRCULAR KNITTING MACHINE

At one feeder, long cylinder and dial needles form stitches. At the next feeder, short cylinder and dial needles form stitches. The needles not knitting at a feeder move through a non-knitting track. Modern machines usually use needles of the same length.



CAM SYSTEM OF INTERLOCK KNITTING MACHINE

Cylinder cam system:

The cylinder needle cam system consists of the followings:

- ❖ A, is a clearing cam which lifts the needle to clear the old loops.
- ❖ B and C, are the stitch and guard cams respectively and are vertically adjustable for varying stitch length.
- ❖ D, is upthrow cam, to raise cylinder needle whilst dial needle knocks over.
- ❖ E and F, are the guard cams, to complete the track.
- ❖ G and H, provide the track for the idling needles.

Dial cam system:

The dial needle cam system consists of the followings:

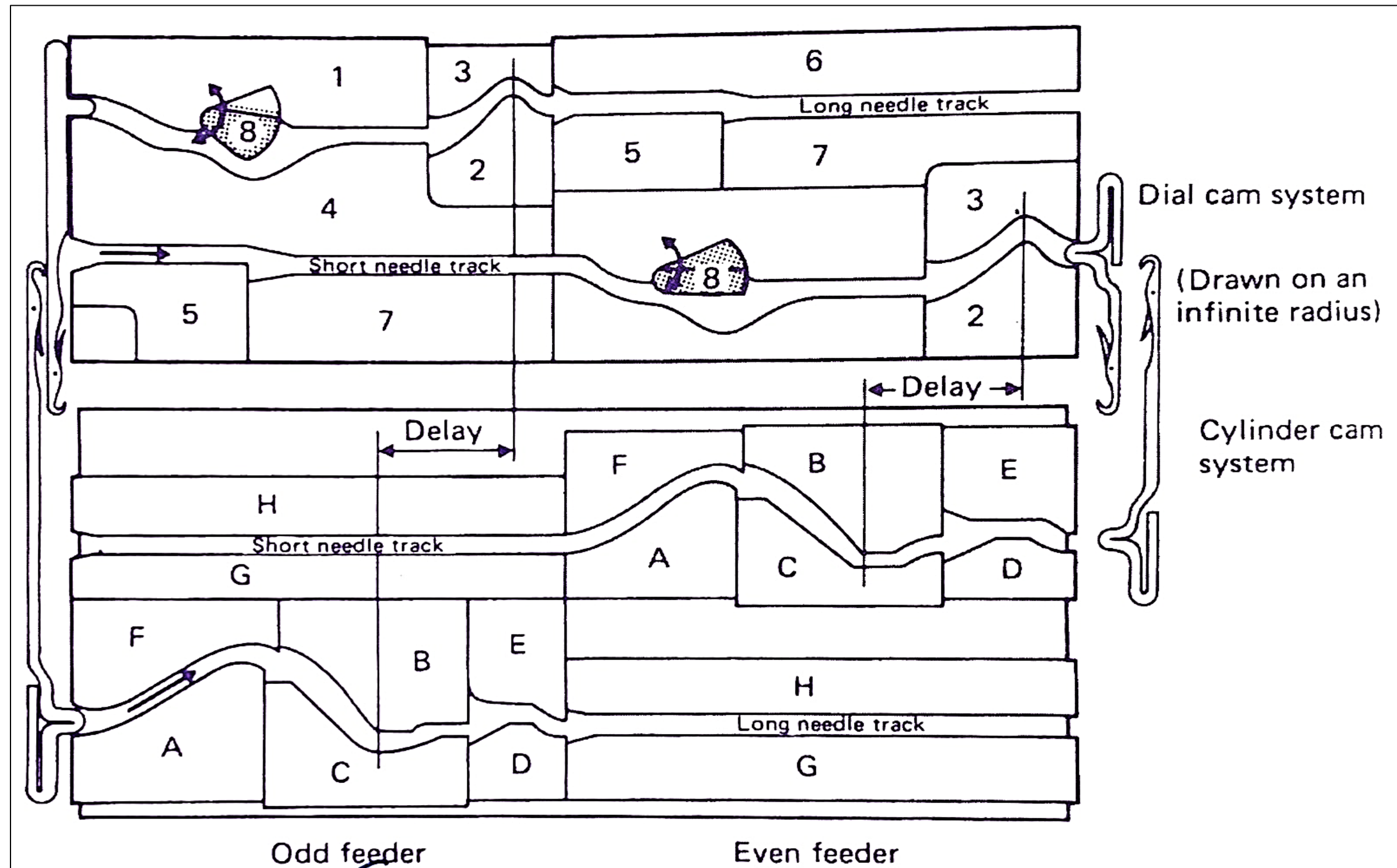
CAM SYSTEM OF INTERLOCK KNITTING MACHINE

Dial cam system:

The dial needle cam system consists of the followings:

- ❖ 1, is a raising cam to the tuck position only.
- ❖ 2 and 3, are the adjustable dial knock-over cams.
- ❖ 4, is a guard cam to complete the track. (Drawn on an infinite radius) Cylinder cam system
- ❖ 5, is an auxiliary knock-over cam to prevent the dial needle re-entering old loop.
- ❖ 6 and 7, provide the track for the idling needles.
- ❖ 8, is a swing type clearing cam, which may occupy the knitting position as shown at feeder 1 or the tuck position as shown at feeder 2.

CAM SYSTEM OF INTERLOCK KNITTING MACHINE



PURL KNITTING MACHINE

In a flat links-links machine, double-headed latch needles are used. These needles can move between two needle beds because the slots of both beds are placed opposite each other.

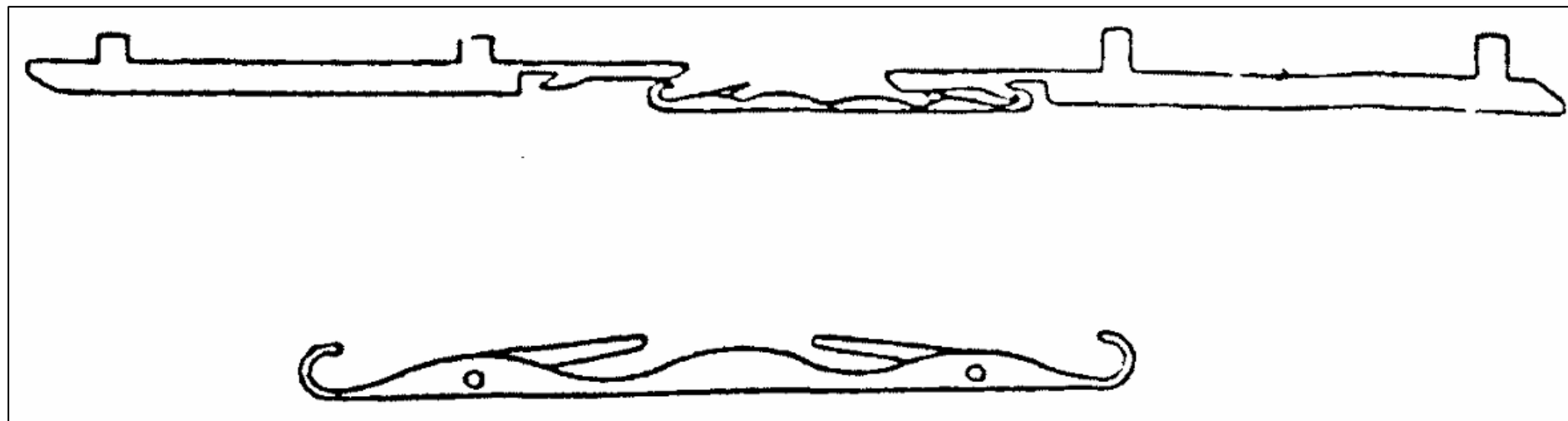
The needle beds have special knock-over teeth that hold the old loops during stitch formation.

Each needle has two sliders:

Slider M controls the left head of the needle.

Slider N controls the right head of the needle.

These sliders move the needle back and forth to form stitches.



PURL KNITTING MACHINE

Stitch Formation Process

1. Needle Movement and Loop Transfer

- ☐ A slider moves the needle from one bed to the other.
- ☐ The old loop opens the needle latch and moves to the middle of the needle.

2. Yarn Laying

- ☐ The yarn is placed on the needle hook.
- ☐ The yarn must pass under the hook correctly and should not be caught by the latch.

3. Underlapping and Pressing

- ☐ The needle moves forward.
- ☐ The old loop is held by the knock-over teeth.
- ☐ The old loop closes the latch, trapping the new yarn inside the needle hook.

PURL KNITTING MACHINE

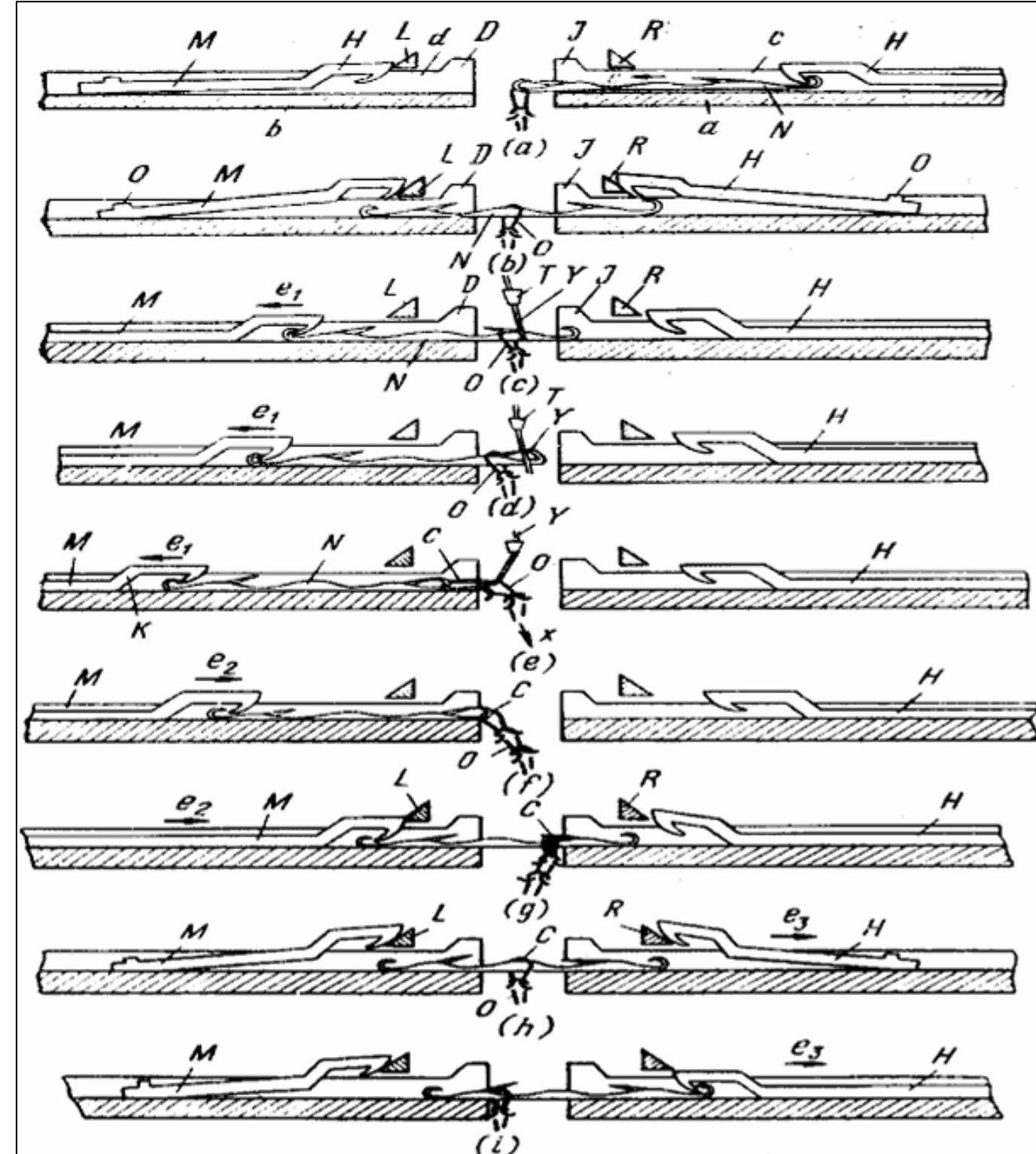
4. Landing, Joining, and Casting-off

- ☐ The needle continues moving.
- ☐ The old loop moves over the latch and joins with the new yarn.
- ☐ The old loop is removed, and a new loop is formed.

5. Formation of Purl Stitch

- ☐ When the left needle head forms the loop, the old loop is cast off to the left side.
- ☐ When the right needle head forms the loop, the old loop is cast off to the right side.
- ☐ This opposite loop formation creates a purl stitch.

PURL KNITTING MACHINE



THANK
YOU !

